

P-BLOCK MASTER STATEMENT

ASSIGNMENT

GROUP 16 - OXYGEN FAMILY

PART I — GROUP 16

Q1. Occurrence and natural abundance [Count the correct statements.]

- (1) Oxygen, sulphur, selenium, tellurium and polonium constitute Group 16.
- (2) Group 16 is also known as the chalcogen family; the name is derived from a Greek word for brass and points to the association of sulphur and its congeners with copper.
- (3) Oxygen is the most abundant of all the elements on earth and forms about 46.6% by mass of the earth's crust.
- (4) Dry air contains about 20.946% oxygen by volume.
- (5) The abundance of sulphur in the earth's crust is only about 0.03-0.1%.
- (6) Gypsum, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, is an important naturally occurring sulphate.
- (7) Epsom salt, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, is a sulphur-containing mineral.
- (8) Baryte has the formula BaSO_4 .
- (9) Galena, zinc blende and copper pyrites are represented by PbS , ZnS and CuFeS_2 respectively.
- (10) Traces of sulphur occur as hydrogen sulphide in volcanoes.
- (11) Eggs, proteins, garlic, onion, mustard, hair and wool contain sulphur.
- (12) Selenium and tellurium occur as metal selenides and tellurides in sulphide ores.
- (13) Polonium occurs in nature as a decay product of thorium and uranium minerals.
- (14) Sulphur is more abundant than oxygen in the earth's crust.
- (15) Most copper minerals contain either oxygen or sulphur and frequently contain other members of the group as well.

Answer: _____

Q2. Electronic configuration, size and ionisation enthalpy [Count the correct statements.]

- (1) The general valence-shell electronic configuration of Group 16 is ns^2np^4 .
- (2) The Group 16 elements have six electrons in their outermost shell.
- (3) Atomic radii generally increase from oxygen to polonium.
- (4) Ionic radii of E^{2-} ions generally increase down the group.
- (5) The oxygen atom is exceptionally small compared with the heavier members.
- (6) First ionisation enthalpy generally decreases down the group.
- (7) The decrease in ionisation enthalpy down the group is mainly related to increasing atomic size.
- (8) Group 16 elements have lower first ionisation enthalpies than the corresponding Group 15 elements.
- (9) The extra stability of the half-filled p^3 configuration helps explain the higher ionisation enthalpy of Group 15.
- (10) The first ionisation enthalpy increases regularly from O to Po.
- (11) Oxygen has the largest atomic radius in Group 16.
- (12) Group 16 elements have higher first ionisation enthalpies than Group 15 elements in the same periods.
- (13) The general valence-shell electronic configuration is ns^2np^5 .

(14) Increasing number of electron shells is responsible for the increase in size down the group.

(15) The ionic radius of O^{2-} is larger than that of Te^{2-} .

Answer: _____

Q3. EXCLUSIVE ORDER QUESTION - periodic properties [Count the correct statements.]

(1) Atomic radius: $O < S < Se < Te < Po$.

(2) E^{2-} ionic radius: $O^{2-} < S^{2-} < Se^{2-} < Te^{2-} < Po^{2-}$.

(3) First ionisation enthalpy: $O > S > Se > Te > Po$.

(4) Electronegativity: $O > S > Se > Te > Po$.

(5) Electron gain enthalpy is most negative for sulphur among O, S, Se, Te and Po.

(6) From S to Po, electron gain enthalpy becomes progressively less negative.

(7) Metallic character: $O < S < Se < Te < Po$.

(8) Stability of the -2 oxidation state decreases down the group.

(9) Stability of the +6 oxidation state decreases down the group.

(10) Stability of the +4 oxidation state increases down the group.

(11) Acidic character of H_2E : $H_2O < H_2S < H_2Se < H_2Te$.

(12) Thermal stability of H_2E decreases on moving down the group.

(13) Reducing character of H_2S , H_2Se and H_2Te increases down the group.

(14) H-E bond dissociation enthalpy: $H-O > H-S > H-Se > H-Te$.

(15) Electron gain enthalpy becomes steadily more negative from O to Po.

Answer: _____

Q4. Atomic and physical data - Data mastery I [Count the correct statements.]

(1) Atomic numbers of O, S, Se, Te and Po are 8, 16, 34, 52 and 84 respectively.

(2) Approximate atomic masses for O, S, Se, Te and Po are 16.00, 32.06, 78.96, 127.60 and 210.00 g mol⁻¹.

(3) Covalent radii for O, S, Se, Te and Po are 66, 104, 117, 137 and 146 pm respectively.

(4) E^{2-} ionic radii for O, S, Se, Te and Po are 140, 184, 198, 221 and about 230 pm respectively.

(5) Electron gain enthalpy values for O and S are about -141 and -200 kJ mol⁻¹ respectively.

(6) Electron gain enthalpy values for Se, Te and Po are about -195, -190 and -174 kJ mol⁻¹ respectively.

(7) First ionisation enthalpies for O and S are about 1314 and 1000 kJ mol⁻¹.

(8) First ionisation enthalpies for Se, Te and Po are about 941, 869 and 813 kJ mol⁻¹.

(9) The covalent radius of oxygen is 104 pm.

(10) The electron gain enthalpy of oxygen is more negative than that of sulphur.

(11) The first ionisation enthalpy of polonium exceeds that of oxygen.

(12) The electronegativities are approximately O 3.50, S 2.58, Se 2.55, Te 2.01 and Po 1.76.

(13) Tellurium is as more electronegative than selenium.

(14) The numerical data show a much larger size jump from O to S than between several successive heavier members.

(15) The electron gain enthalpy of polonium is as -274 kJ mol⁻¹.

Answer: _____

Q5. Physical properties, allotropy and atomicity [Count the correct statements.]

(1) Oxygen and sulphur are non-metals.

(2) Selenium and tellurium are described as metalloids.

(3) Polonium is metallic.

(4) Polonium is radioactive and short lived.

(5) All Group 16 elements exhibit allotropy.

(6) Melting and boiling points generally increase with atomic number down the group.

(7) A large difference between oxygen and sulphur melting/boiling points is related to their different atomicities.

(8) Elemental oxygen exists as O_2 whereas common sulphur exists as S_8 .

(9) Oxygen exists as O_8 in its ordinary elemental state.

(10) Sulphur is a diatomic molecule under ordinary conditions.

(11) Polonium is described as a non-metal.

(12) Selenium and tellurium are described as typical metals.

(13) The melting point of oxygen is about 55 K.

(14) The boiling point of oxygen is about 90 K.

(15) The boiling point of sulphur is lower than that of oxygen.

Answer: _____

Q6. Oxidation states and inert-pair trend [Count the correct statements.]

(1) The stability of the -2 oxidation state decreases down Group 16.

(2) Polonium hardly shows the -2 oxidation state.

(3) Oxygen generally shows the -2 oxidation state in its compounds.

(4) Oxygen has positive oxidation states in compounds with fluorine.

(5) The oxidation state of oxygen in OF_2 is +2.

(6) The oxidation state of oxygen in O_2F_2 is +1 per oxygen atom.

(7) Sulphur, selenium and tellurium commonly show +4 in compounds with oxygen.

(8) Sulphur, selenium and tellurium can show +6, especially in compounds with fluorine.

(9) The +6 oxidation state becomes less stable down the group.

(10) The +4 oxidation state becomes more stable down the group because of the inert pair effect.

(11) Bonding in the +4 and +6 oxidation states is primarily covalent.

(12) The +6 oxidation state becomes more stable from S to Po.

(13) Oxygen commonly expands its valence shell to show +6.

(14) The +4 oxidation state becomes less stable down the group.

(15) Polonium most commonly exhibits a stable -2 state according to the trend.

Answer: _____

Q7. Anomalous behaviour of oxygen [Count the correct statements.]

(1) The anomalous behaviour of oxygen is related to its very small size.

(2) Its high electronegativity contributes to the anomalous behaviour.

(3) Strong hydrogen bonding in H_2O is an important consequence of oxygen's small size and high electronegativity.

(4) Comparable strong hydrogen bonding is not present in H_2S .

(5) Oxygen has no valence-shell d orbitals.

(6) Absence of d orbitals limits the covalency of oxygen.

(7) In practice, the covalency of oxygen rarely exceeds two.

(8) Heavier Group 16 elements can expand their valence shells so that covalency can exceed four.

(9) Oxygen readily uses 2d orbitals for expansion of its valence shell.

(10) H_2S has stronger hydrogen bonding than H_2O .

(11) The anomalous behaviour of oxygen is due mainly to its very large size.

(12) Oxygen is the second most electronegative element after fluorine.

(13) Oxygen has the lowest electronegativity of Group 16.

- (14) The compactness of oxygen also explains why its electron gain enthalpy is less negative than that of sulphur.
- (15) The electron gain enthalpy of oxygen is more negative than that of sulphur because oxygen is smaller.

Answer: _____

Q8. Hydrides H₂E - trends and chemistry [Count the correct statements.]

- (1) All Group 16 elements form hydrides of the type H₂E.
- (2) Acidic character increases from H₂O to H₂Te.
- (3) The increase in acidity down the group is related to decreasing H-E bond dissociation enthalpy.
- (4) Thermal stability decreases from H₂O down the group.
- (5) All hydrides except water possess reducing character.
- (6) Reducing character increases from H₂S to H₂Te.
- (7) H₂Te is more acidic than H₂S.
- (8) H₂Te is a stronger reducing agent than H₂S.
- (9) H₂O is the strongest reducing hydride of the group.
- (10) Thermal stability increases from H₂O to H₂Te.
- (11) H₂S is more acidic than H₂Te.
- (12) The H-E bond length increases from H₂O to H₂Te.
- (13) The H-E-H bond angle decreases toward 90° on moving down the series.
- (14) The anomalously high boiling point of H₂O is associated with hydrogen bonding.
- (15) H₂O has the weakest H-E bond in the series H₂O, H₂S, H₂Se, H₂Te.

Answer: _____

Q9. Hydride data - exact numerical trends [Count the correct statements.]

- (1) The melting points of H₂O, H₂S, H₂Se and H₂Te are about 273, 188, 208 and 222 K.
- (2) The boiling points of H₂O, H₂S, H₂Se and H₂Te are about 373, 213, 232 and 269 K.
- (3) The H-E distances for H₂O, H₂S, H₂Se and H₂Te are about 96, 134, 146 and 169 pm.
- (4) The H-E-H angles are about 104°, 92°, 91° and 90° respectively.
- (5) The H-E bond dissociation enthalpies are about 463, 347, 276 and 238 kJ mol⁻¹.
- (6) The first dissociation constants increase strongly from H₂O to H₂Te.
- (7) H₂O has the highest boiling point in the hydride series.
- (8) H₂S has the lowest boiling point among the four hydrides.
- (9) H₂Te has an H-E bond distance smaller than H₂S.
- (10) The H-E-H angle for H₂O is about 90°.
- (11) The bond dissociation enthalpy of H-Te is greater than H-O.
- (12) The numerical data support decreasing thermal stability down the group.
- (13) The numerical dissociation constants support increasing acidity down the group.
- (14) The boiling point of H₂Se exceeds that of H₂Te.
- (15) The H-E bond length decreases from H₂O to H₂Te.

Answer: _____

Q10. Oxides of Group 16 [Count the correct statements.]

- (1) The heavier Group 16 elements form oxides of the EO₂ and EO₃ types.
- (2) Ozone, O₃, and sulphur dioxide, SO₂, are gases under ordinary conditions discussed.
- (3) Selenium dioxide, SeO₂, is a solid.
- (4) The reducing property of the dioxides decreases from SO₂ to TeO₂.
- (5) SO₂ behaves as a reducing agent.
- (6) TeO₂ is described as an oxidising agent in this trend.
- (7) Sulphur, selenium and tellurium also form EO₃ type oxides: SO₃, SeO₃ and TeO₃.
- (8) Both EO₂ and EO₃ oxide types are described as acidic.
- (9) Reducing character of the dioxides increases from SO₂ to TeO₂.

- (10) TeO₂ is a stronger reducing agent than SO₂.
- (11) SeO₂ is described as a gas at room temperature.
- (12) SO₃ is described as a basic oxide.
- (13) The +4 oxidation state is represented in EO₂ type oxides such as SO₂.
- (14) The +6 oxidation state is represented in EO₃ type oxides such as SO₃.
- (15) Oxygen forms a stable EO₃ oxide of itself in the same sense as SO₃.

Answer: _____

Q11. Halides of Group 16 [Count the correct statements.]

- (1) Group 16 elements form many halides of the types EX₆, EX₄ and EX₂.
- (2) The stability of halides is described as decreasing in the order fluoride > chloride > bromide > iodide.
- (3) Among hexahalides, the hexafluorides are the stable members.
- (4) The hexafluorides are gaseous and have octahedral structures.
- (5) SF₆ is exceptionally stable for steric reasons.
- (6) SF₄ is a gas, SeF₄ a liquid and TeF₄ a solid.
- (7) The tetrafluorides are described with a trigonal-bipyramidal electron arrangement and see-saw molecular geometry.
- (8) One equatorial position in the tetrafluoride arrangement is occupied by a lone pair.
- (9) All Group 16 elements except oxygen form dichlorides and dibromides.
- (10) The well-known monohalides are dimeric, such as S₂F₂, S₂Cl₂ and Se₂Cl₂.
- (11) Dimeric halides can undergo disproportionation.
- (12) One disproportionation is 2Se₂Cl₂ → SeCl₄ + 3Se.
- (13) SF₆ has a tetrahedral structure.
- (14) TeF₄ is a gas.
- (15) Oxygen readily forms stable dichlorides and dibromides of the same type as sulphur.

Answer: _____

Q12. Other sulphur species and paramagnetism [Count the correct statements.]

- (1) Several ring modifications containing about 6-20 sulphur atoms have been synthesised.
- (2) Cyclo-S₆ adopts a chair form.
- (3) At elevated temperatures near 1000 K, S₂ becomes the dominant sulphur species.
- (4) S₂ is paramagnetic like O₂.
- (5) The paramagnetism of S₂ is associated with two unpaired electrons in antibonding π* orbitals.
- (6) S₂ is diamagnetic because all electrons are paired.
- (7) Cyclo-S₆ is described as planar hexagonal.
- (8) The atomicity of sulphur changes with temperature.
- (9) Paramagnetic behaviour is specifically associated with sulphur vapour containing S₂.
- (10) Only S₈ can exist in sulphur vapour at any temperature.
- (11) Rhombic and monoclinic forms differ in crystal packing even though both contain S₈ molecules.
- (12) The α and β forms differ because one contains S₈ and the other S₆.
- (13) The S₈ structural unit in the common allotropes is non-planar.
- (14) The transition between α and β sulphur changes the molecular formula from S₈ to S₂.
- (15) The high-temperature S₂ species provides an analogy with O₂ in magnetic behaviour.

Answer: _____

Q13. Advanced oxoacid acidity, oxidation-state and linkage challenge [Count the correct statements.]

- (1) For oxoacids of the same central atom, increasing number of terminal oxo groups generally increases acidity.

- (2) H_2SO_4 is a stronger acid than H_2SO_3 in aqueous solution.
- (3) A peroxo O-O linkage does not require sulphur to have an oxidation state above +6.
- (4) In H_2SO_5 , the oxidation state of sulphur is +6.
- (5) In $\text{H}_2\text{S}_2\text{O}_8$, each sulphur is in the +6 oxidation state.
- (6) In $\text{H}_2\text{S}_2\text{O}_7$, the S-O-S bridge is not a peroxide linkage.
- (7) In a peroxide linkage each oxygen is assigned oxidation state -1 for oxidation-number calculation.
- (8) Sulphur reaches +7 in $\text{H}_2\text{S}_2\text{O}_8$ because the molecule contains eight oxygen atoms.
- (9) Every O-O bond in an oxoacid is assigned oxygen oxidation state -2.
- (10) The presence of O-O versus S-O-S linkages is a key structural distinction among sulphur oxoacids.
- (11) Oxidation-number assignment must be based on bonding/charge rules rather than simply counting oxygen atoms.
- (12) H_2SO_3 and H_2SO_4 contain sulphur in the same oxidation state.
- (13) $\text{H}_2\text{S}_2\text{O}_7$ can be formed by absorption of SO_3 in H_2SO_4 .
- (14) Dilution of $\text{H}_2\text{S}_2\text{O}_7$ produces H_2SO_4 .
- (15) H_2SO_4 contains an O-O peroxide bond.

Answer: _____

Q14. Group 16 reaction-equation mastery [Count the correct statements.]

- (1) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$ is a preparation of dioxygen.
- (2) $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ is a catalytic preparation of dioxygen.
- (3) $3\text{O}_2 \rightarrow 2\text{O}_3$ represents ozone formation.
- (4) $\text{PbS} + 4\text{O}_3 \rightarrow \text{PbSO}_4 + 4\text{O}_2$ illustrates ozone oxidation.
- (5) $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$ is the direct preparation of sulphur dioxide.
- (6) $\text{SO}_3^{2-} + 2\text{H}^+ \rightarrow \text{H}_2\text{O} + \text{SO}_2$ is a laboratory generation of SO_2 .
- (7) $2\text{NaOH} + \text{SO}_2 \rightarrow \text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$ occurs with alkali.
- (8) $\text{Na}_2\text{SO}_3 + \text{H}_2\text{O} + \text{SO}_2 \rightarrow 2\text{NaHSO}_3$ occurs with further SO_2 .
- (9) $\text{SO}_2 + \text{Cl}_2 \rightarrow \text{SO}_2\text{Cl}_2$ is catalysed by charcoal.
- (10) $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ is the key Contact-process oxidation.
- (11) $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$ forms oleum.
- (12) $\text{Cu} + 2\text{H}_2\text{SO}_4(\text{conc.}) \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$.
- (13) $\text{S} + 2\text{H}_2\text{SO}_4(\text{conc.}) \rightarrow 3\text{SO}_2 + 2\text{H}_2\text{O}$.
- (14) $\text{C} + 2\text{H}_2\text{SO}_4(\text{conc.}) \rightarrow \text{CO}_2 + 2\text{SO}_2 + 2\text{H}_2\text{O}$.
- (15) $\text{SO}_3 + \text{H}_2\text{SO}_4$ directly gives H_2SO_3 .

Answer: _____

Q15. Atomic and physical data - Data mastery II [Count the correct statements.]

- (1) The densities given for O, S, Se and Te are approximately 1.32, 2.06, 4.19 and 6.25 g cm⁻³ respectively under the specified conditions.
- (2) The sulphur density value 2.06 g cm⁻³ corresponds to rhombic sulphur.
- (3) The selenium density value is associated with hexagonal grey selenium.
- (4) Melting points for O, S, Se, Te and Po are approximately 55, 393, 490, 725 and 520 K.
- (5) Boiling points for O, S, Se, Te and Po are approximately 90, 718, 958, 1260 and 1235 K.
- (6) The melting point does not increase monotonically from Te to Po; Po is lower than Te.
- (7) The boiling point of Po is lower than that of Te.
- (8) Oxygen commonly exhibits oxidation states -2 and -1 and positive states +1 and +2 in oxygen fluorides.
- (9) Sulphur, selenium and tellurium show -2, +2, +4 and +6 oxidation states.
- (10) Polonium is represented mainly by +2 and +4 oxidation states in the data table.
- (11) Polonium has a short half-life of about 13.8 days.
- (12) Polonium has a higher boiling point than tellurium in the data table.

- (13) The melting point of oxygen is about 393 K.

- (14) Sulphur is represented only by -2 and +6 oxidation states.

- (15) The physical data show a general rise in melting and boiling points down the group with notable end-of-group deviations.

Answer: _____

Q16. Hydride data - formation enthalpy and dissociation constant [Count the correct statements.]

- (1) Standard enthalpies of formation for H_2O , H_2S , H_2Se and H_2Te are about -286, -20, +73 and +100 kJ mol⁻¹ respectively.
- (2) The enthalpy of formation becomes progressively less favourable from H_2O to H_2Te .
- (3) The first dissociation constant of H_2O is about 1.8×10^{-16} at 298 K.
- (4) The first dissociation constant of H_2S is about 1.3×10^{-7} at 298 K.
- (5) The first dissociation constant of H_2Se is about 1.3×10^{-4} at 298 K.
- (6) The first dissociation constant of H_2Te is about 2.3×10^{-3} at 298 K.
- (7) The dissociation constants increase by many orders of magnitude from H_2O to H_2Te .
- (8) The numerical K_a values are consistent with H_2Te being much more acidic than H_2S .
- (9) The H-E bond dissociation enthalpy falls from 463 kJ mol⁻¹ for H-O to 238 kJ mol⁻¹ for H-Te.
- (10) The first dissociation constant of H_2O is larger than that of H_2Te .
- (11) H_2S has a positive formation enthalpy of about +100 kJ mol⁻¹.
- (12) H_2Te has the strongest H-E bond among the four hydrides.
- (13) Increasing acidity parallels decreasing H-E bond strength down the group.
- (14) The formation-enthalpy data also reflect decreasing thermochemical stability down the group.
- (15) All four hydrides have negative standard enthalpies of formation.

Answer: _____

Q17. ALL USES - Group 16 complete master coverage [Count the correct statements.]

- (1) Dioxygen is essential in normal respiration and supports combustion processes.
- (2) Oxygen is used in oxyacetylene welding.
- (3) Oxygen is used in the manufacture of many metals, particularly steel.
- (4) Oxygen cylinders are used in hospitals, high-altitude flying and mountaineering.
- (5) Liquid oxygen is used with fuels such as hydrazines to provide very high thrust in rockets.
- (6) Ozone is used as a germicide and disinfectant.
- (7) Ozone is used for sterilising water.
- (8) Ozone is used for bleaching oils, ivory, flour and starch.
- (9) Ozone is used as an oxidising agent in the manufacture of potassium permanganate.
- (10) Sulphur dioxide is used in refining petroleum and sugar.
- (11) Sulphur dioxide is used for bleaching wool and silk and also as an anti-chlor, disinfectant and preservative.
- (12) Sulphur dioxide is used to manufacture sulphuric acid, sodium hydrogen sulphite and calcium hydrogen sulphite.
- (13) Liquid sulphur dioxide is used as a solvent for a number of organic and inorganic chemicals.
- (14) Major sulphuric-acid uses include fertilisers, petroleum refining, pigments, paints, dyestuff intermediates, detergents, metallurgical cleaning, storage batteries, nitrocellulose products and laboratory work.
- (15) Ozone, sulphur dioxide and sulphuric acid have no important industrial or laboratory uses.

Answer: _____

Q18. Dioxygen - fine preparation, physical data and exceptions [Count the correct statements.]

- (1) Heating KClO_3 in the presence of MnO_2 is a laboratory method for obtaining dioxygen.
- (2) Oxygen-containing salts such as chlorates, nitrates and permanganates can furnish dioxygen on heating.
- (3) Thermal decomposition of Ag_2O can be represented as $2\text{Ag}_2\text{O} \rightarrow 4\text{Ag} + \text{O}_2$.
- (4) Heating Pb_3O_4 can give PbO and O_2 according to $2\text{Pb}_3\text{O}_4 \rightarrow 6\text{PbO} + \text{O}_2$.
- (5) Thermal decomposition of HgO gives mercury and dioxygen.
- (6) PbO_2 can decompose on heating to PbO and O_2 .
- (7) H_2O_2 readily decomposes to H_2O and O_2 in the presence of catalysts such as finely divided metals or MnO_2 .
- (8) During electrolysis of water, oxygen is liberated at the cathode.
- (9) Industrial dioxygen can be obtained by removing CO_2 and water vapour from air, liquefying the remaining gases and fractionally distilling them.
- (10) Dioxygen is a colourless and odourless gas.
- (11) At 293 K, about 3.08 cm^3 of O_2 dissolves in 100 cm^3 of water.
- (12) Dioxygen liquefies at about 90 K and freezes at about 55 K.
- (13) Oxygen has three stable isotopes: ^{16}O , ^{17}O and ^{18}O .
- (14) Molecular oxygen is diamagnetic despite having an even number of electrons.
- (15) Dioxygen reacts directly with nearly all metals and non-metals except some species such as Au, Pt and some noble gases.

Answer: _____

Q19. Dioxygen - equations, bond enthalpy and complete uses audit [Count the correct statements.]

- (1) Calcium burns in oxygen according to $2\text{Ca} + \text{O}_2 \rightarrow 2\text{CaO}$.
- (2) Aluminium reacts with oxygen according to $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$.
- (3) Phosphorus burns in oxygen to form P_4O_{10} .
- (4) Carbon burns in oxygen to form CO_2 .
- (5) Roasting ZnS in oxygen gives ZnO and SO_2 .
- (6) Complete combustion of methane can be written $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$.
- (7) Catalytic oxidation of SO_2 by O_2 in the presence of V_2O_5 gives SO_3 .
- (8) In the presence of CuCl_2 , O_2 can oxidise HCl to Cl_2 and H_2O .
- (9) The $\text{O}=\text{O}$ bond dissociation enthalpy is about $493.4 \text{ kJ mol}^{-1}$.
- (10) External heating is often required to initiate reactions with dioxygen because of the high $\text{O}=\text{O}$ bond dissociation enthalpy.
- (11) Dioxygen is used in oxyacetylene welding.
- (12) Dioxygen is used in manufacture of metals, particularly steel.
- (13) Oxygen cylinders are used in hospitals, high-altitude flying and mountaineering.
- (14) Combustion of fuels such as hydrazines in liquid oxygen can provide tremendous thrust in rockets.
- (15) Platinum reacts rapidly and directly with dioxygen under ordinary conditions.

Answer: _____

Q20. Simple oxides - complete classification and examples [Count the correct statements.]

- (1) A binary compound of oxygen with another element is called an oxide.
- (2) MgO and Al_2O_3 are examples of simple oxides.
- (3) Pb_3O_4 and Fe_3O_4 are examples of mixed oxides.
- (4) An oxide that combines with water to give an acid is termed an acidic oxide.

- (5) SO_2 , Cl_2O_7 , CO_2 and N_2O_5 are examples of acidic oxides.
- (6) SO_2 combines with water to give H_2SO_3 .
- (7) Only non-metal oxides can possess acidic character.
- (8) Mn_2O_7 , CrO_3 and V_2O_5 are metal oxides in high oxidation states that show acidic character.
- (9) An oxide that gives a base with water is called a basic oxide.
- (10) Na_2O , CaO and BaO are examples of basic oxides.
- (11) $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$ illustrates formation of a base from a basic oxide.
- (12) Metallic oxides are generally basic.
- (13) Al_2O_3 is amphoteric and reacts with acids as well as alkalies.
- (14) CO , NO and N_2O are examples of neutral oxides.
- (15) Neutral oxides necessarily react with water to form strong acids.

Answer: _____

Q21. Ozone - exact preparation, numerical data and thermodynamic stability [Count the correct statements.]

- (1) Ozone is an allotropic form of oxygen.
- (2) Ozone is too reactive to remain for long in the atmosphere at sea level.
- (3) At a height of about 20 km, ozone is formed from atmospheric oxygen in the presence of sunlight.
- (4) The ozone layer protects the earth from excessive ultraviolet radiation.
- (5) Passing a slow dry stream of oxygen through a silent electrical discharge gives about 10% ozone in ozonised oxygen.
- (6) For $3\text{O}_2 \rightarrow 2\text{O}_3$, $\Delta H^\circ(298 \text{ K})$ is approximately $+142 \text{ kJ mol}^{-1}$.
- (7) Formation of ozone from oxygen is endothermic.
- (8) A silent electrical discharge is used so that excessive heating does not promote decomposition of ozone.
- (9) A battery of ozonisers can be used when ozone concentrations greater than 10% are required.
- (10) Pure ozone has a boiling point close to 101.1 K.
- (11) Pure ozone is a pale blue gas, a dark blue liquid and a violet-black solid.
- (12) Ozone has a characteristic smell.
- (13) Above about 100 parts per million, breathing ozone becomes uncomfortable and may cause headache and nausea.
- (14) Decomposition of ozone to dioxygen is endothermic.
- (15) For conversion of ozone to oxygen, ΔH is negative and ΔS is positive, making the process thermodynamically favourable.

Answer: _____

Q22. Ozone - analysis, atmospheric chemistry, structure and uses [Count the correct statements.]

- (1) $\text{O}_3 \rightarrow \text{O}_2 + \text{O}$ is used to represent the ready liberation of nascent oxygen by ozone.
- (2) Lead sulphide is oxidised by ozone to lead sulphate.
- (3) Ozone oxidises iodide ions to iodine in aqueous medium.
- (4) For quantitative estimation, ozone is passed into excess KI solution buffered with borate buffer at about pH 9.2.
- (5) The iodine liberated in ozone estimation can be titrated with standard sodium thiosulphate solution.
- (6) The iodometric method provides a quantitative estimate of ozone.
- (7) Nitric oxide reacts rapidly with ozone to form NO_2 and O_2 .
- (8) Nitrogen oxides from supersonic jet exhausts have been considered a possible cause of depletion of upper-atmosphere ozone.
- (9) Freons used in aerosol sprays and as refrigerants are another threat to the ozone layer.

- (10) The two O-O bond lengths in ozone are identical and are about 128 pm.
 (11) The O-O-O bond angle in ozone is about 117°.
 (12) Ozone is a linear molecule.
 (13) Ozone is a resonance hybrid of two principal canonical forms.
 (14) Ozone is used as a germicide, disinfectant and for sterilising water.
 (15) Ozone is used for bleaching oils, ivory, flour and starch and as an oxidising agent in manufacture of potassium permanganate.

Answer: _____

Q23. Sulphur allotropes - exact melting points, densities, preparation and structure [Count the correct statements.]

- (1) Rhombic sulphur is the stable form at room temperature.
 (2) Rhombic sulphur transforms into monoclinic sulphur above 369 K.
 (3) Rhombic sulphur is yellow, melts at about 385.8 K and has specific gravity about 2.06.
 (4) Rhombic sulphur crystals can be obtained by evaporating a solution of roll sulphur in CS₂.
 (5) Rhombic sulphur is insoluble in water, dissolves to some extent in benzene, alcohol and ether, and is readily soluble in CS₂.
 (6) Monoclinic sulphur melts at about 393 K and has specific gravity about 1.98.
 (7) Monoclinic sulphur can be prepared by melting rhombic sulphur, allowing a crust to form, making holes and pouring out the remaining liquid.
 (8) Monoclinic sulphur forms colourless needle-shaped crystals.
 (9) Monoclinic sulphur is stable above 369 K and changes to rhombic sulphur below 369 K.
 (10) At 369 K both rhombic and monoclinic forms are stable; this is the transition temperature.
 (11) Both rhombic and monoclinic sulphur contain S₈ molecules.
 (12) The S₈ ring in these forms is planar.
 (13) Ring allotropes containing roughly 6-20 sulphur atoms have been synthesised.
 (14) Cyclo-S₆ adopts a chair form.
 (15) At temperatures around 1000 K, S₂ becomes a dominant vapour species and is paramagnetic like O₂.

Answer: _____

Q24. Sulphur dioxide - fine preparation, physical constants, reactions and uses [Count the correct statements.]

- (1) Burning sulphur in air or oxygen gives SO₂ together with a small amount, about 6-8%, of SO₃.
 (2) In the laboratory, SO₂ can be generated by treating a sulphite with dilute sulphuric acid.
 (3) Industrial SO₂ is produced as a by-product during roasting of sulphide ores.
 (4) After drying, SO₂ can be liquefied under pressure and stored in steel cylinders.
 (5) SO₂ is a colourless gas with a pungent smell and is highly soluble in water.
 (6) SO₂ can be liquefied at room temperature under a pressure of about two atmospheres.
 (7) The boiling point of SO₂ is about 263 K.
 (8) Passing SO₂ through water gives a solution of sulphurous acid.
 (9) SO₂ reacts with NaOH first to form sodium sulphite; additional SO₂ gives sodium hydrogen sulphite.
 (10) In its reactions with water and alkalis, SO₂ behaves similarly to CO₂.
 (11) In the presence of charcoal, SO₂ reacts with chlorine to form sulphonyl chloride, SO₂Cl₂.
 (12) Oxygen oxidises SO₂ to SO₃ in the presence of V₂O₅.

- (13) Moist SO₂ acts as a reducing agent, converts Fe(III) to Fe(II) and decolourises acidified permanganate.
 (14) The SO₂ molecule is linear.
 (15) Uses of SO₂ include petroleum and sugar refining, bleaching wool and silk, use as anti-chlor, disinfectant and preservative, manufacture of sulphur chemicals, and use of liquid SO₂ as a solvent.

Answer: _____

Q25. Oxoacids of sulphur - complete formula inventory and stability [Count the correct statements.]

- (1) H₂SO₃ is included among important oxoacids of sulphur.
 (2) H₂S₂O₃ is an oxoacid of sulphur.
 (3) H₂S₂O₄ is included among sulphur oxoacids.
 (4) H₂S₂O₅ is included among sulphur oxoacids.
 (5) Acids of the general formula H₂S_xO₆ with x from 2 to 5 are included in the sulphur oxoacid family.
 (6) H₂SO₄ is an oxoacid of sulphur.
 (7) H₂S₂O₇ is an oxoacid of sulphur.
 (8) H₂SO₅ is an oxoacid of sulphur.
 (9) H₂S₂O₈ is an oxoacid of sulphur.
 (10) Every listed oxoacid of sulphur can be isolated indefinitely as a pure stable substance.
 (11) Some sulphur oxoacids are unstable and are known mainly in aqueous solution or through their salts.
 (12) H₂S₂O₇ is oleum or pyrosulphuric acid in the sulphuric-acid process.
 (13) H₂SO₅ contains a peroxide linkage.
 (14) H₂S₂O₈ contains a peroxide linkage.
 (15) H₂S₂O₆ belongs to the H₂S_xO₆ series for x = 2.

Answer: _____

Q26. Contact process - complete conditions, purification and plant details [Count the correct statements.]

- (1) Sulphuric acid is one of the most important industrial chemicals.
 (2) One step in the Contact process is burning sulphur or sulphide ores in air to generate SO₂.
 (3) SO₂ is oxidised to SO₃ with oxygen in the presence of V₂O₅.
 (4) SO₃ is absorbed in concentrated H₂SO₄ to form oleum.
 (5) The SO₂ feed is purified by removing dust and impurities such as arsenic compounds.
 (6) The key equilibrium is 2SO₂ + O₂ ⇌ 2SO₃ with ΔH° approximately -196.6 kJ mol⁻¹.
 (7) The catalytic oxidation is exothermic, reversible and accompanied by a decrease in gaseous volume.
 (8) Lower temperature and higher pressure favour equilibrium formation of SO₃.
 (9) Extremely low operating temperature is avoided because the reaction rate would become too slow.
 (10) Industrial operation is carried out at about 2 bar and 720 K.
 (11) Industrial SO₃ is preferably absorbed directly in a large quantity of water to avoid oleum formation.
 (12) SO₃ + H₂SO₄ → H₂S₂O₇ is the oleum-forming reaction.
 (13) Dilution of oleum with water gives sulphuric acid of the desired concentration.
 (14) Sulphuric acid obtained by the Contact process is about 96-98% pure.
 (15) The plant flow scheme includes dust removal and an arsenic purifier using gelatinous hydrated ferric oxide.

Answer: _____

Q27. Sulphuric acid - exact physical data, ionisation, volatility, dehydration and uses [Count the correct statements.]

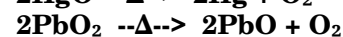
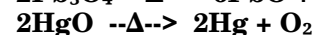
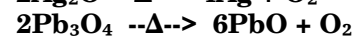
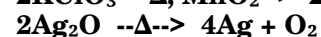
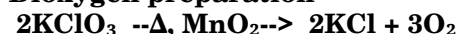
- (1) Sulphuric acid is a colourless, dense, oily liquid with specific gravity about 1.84 at 298 K.
 (2) Its freezing point is about 283 K and boiling point about 611 K.

- (3) Dissolving concentrated H_2SO_4 in water evolves a large quantity of heat.
- (4) For safe dilution, water should be added rapidly to concentrated sulphuric acid.
- (5) Concentrated acid should be added slowly to water with constant stirring.
- (6) Important chemical characteristics include low volatility, strong acidity, strong affinity for water and oxidising ability.
- (7) The first ionisation constant is very large, with $K_{a1} > 10$.
- (8) The second ionisation constant is about 1.2×10^{-2} .
- (9) Sulphuric acid forms normal sulphates as well as acid sulphates.
- (10) Because of low volatility, H_2SO_4 can be used to prepare more volatile acids such as HF, HCl and HNO_3 from their salts.
- (11) Concentrated H_2SO_4 can dry many wet gases provided they do not react with the acid.
- (12) Its dehydrating action chars carbohydrates such as sucrose.
- (13) Hot concentrated H_2SO_4 is a moderately strong oxidising agent and is reduced mainly to SO_2 in the cited reactions.
- (14) Major applications include fertilisers, petroleum refining, pigments/paints/dyestuff intermediates, detergents, metallurgical cleaning, storage batteries, nitrocellulose manufacture and laboratory use.
- (15) Phosphoric acid is a stronger oxidising agent than hot concentrated sulphuric acid.

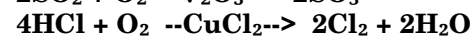
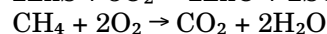
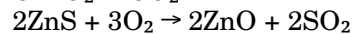
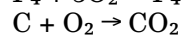
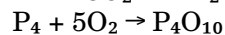
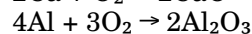
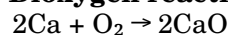
Answer: _____

REACTION BANK — GROUP 16

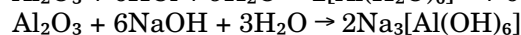
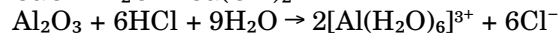
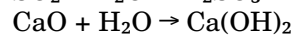
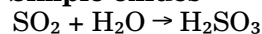
Dioxygen preparation



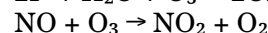
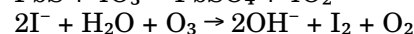
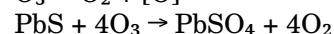
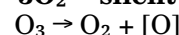
Dioxygen reactions and catalytic oxidations



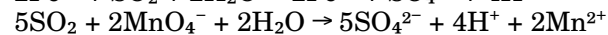
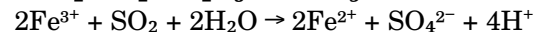
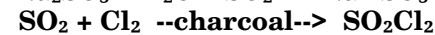
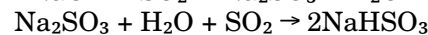
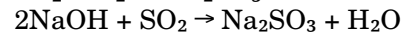
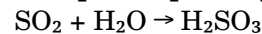
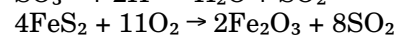
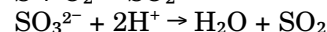
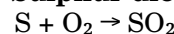
Simple oxides



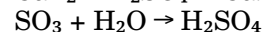
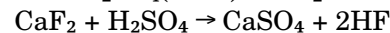
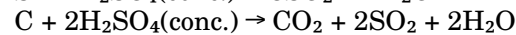
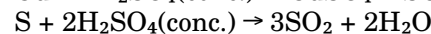
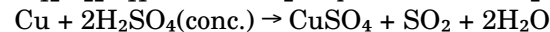
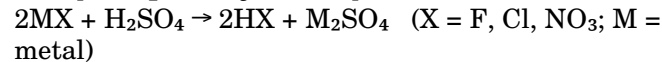
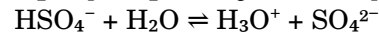
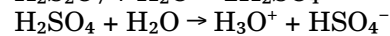
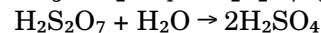
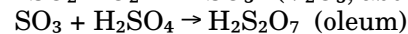
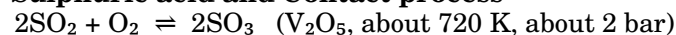
Ozone



Sulphur dioxide



Sulphuric acid and Contact process



ANSWER KEY

PART I — GROUP 16

Q1: 14 Q2: 10 Q3: 14 Q4: 10 Q5: 10 Q6: 11 Q7: 10

Q8: 11 Q9: 10 Q10: 10 Q11: 12 Q12: 10

Q13: 11 Q14: 14 Q15: 12 Q16: 11 Q17: 14 Q18: 13

Q19: 14 Q20: 13 Q21: 14 Q22: 14 Q23: 14 Q24: 14

Q25: 14 Q26: 14 Q27: 13